

Roman Marbel — E-Commerce PRD v1.0

2026-09-19 · @Someone

1. Executive Summary

Roman Marbel is an established marble and tiles retailer at Ausha, Nabastha, Memari, Bardhaman (pin 713407). This document specifies a **mobile-first e-commerce and lead-capture web app** that lets customers across Purba and Paschim Bardhaman select marble, granite and tiles from photographs, describe their room, and receive an accurate quantity-and-price estimate without visiting the shop first.

Why photo-first, not SKU-first. Nobody in this market searches for "Vitrified 600x1200 Glossy Carrara". They walk into a shop, point at a slab, and ask the rate. The app must replicate that: large pictures, tappable, rate visible. Text search is secondary; the photo grid *is* the product.

Why quantity, not just rate. The biggest friction in this trade is the customer not knowing *how much* to buy. They know their room is "12 by 10". They do not know that means 120 sq ft, plus 8% wastage, plus skirting, which works out to 11 boxes of 600x600. Converting room dimensions into a box count and a rupee figure is the core value this app delivers, and it is what turns a browser into a buyer.

Business model. This is not a pure cash-and-carry cart. In tiles retail most orders involve a site visit, a negotiated rate, or a partial advance. The app is therefore designed as **catalogue → estimate → enquiry/quote → confirmed order**, with online payment available but optional. Treating this as a Flipkart clone would fail; treating it as a digital shop-counter that captures qualified leads with quantities already attached will not.

Scale of ambition, stated honestly. MVP is a well-built catalogue with a calculator and a WhatsApp-backed quote flow, shippable in 8 to 10 weeks. AI room visualisation is a Phase 3 differentiator, not a launch requirement. Building the visualiser first is the single most common way projects like this die.

2. Business Context & Market

2.1 The business

Attribute	Detail
Trading name	Roman Marbel

Attribute	Detail
Store address	Ausha, Nabastha, Memari, Bardhaman
PIN	713407
Contact	73836 95415
Category	Marble, granite, vitrified & ceramic tiles
Primary market	Purba Bardhaman & Paschim Bardhaman districts

Note for the owner: the dictated brief mentioned PIN 713401 (Bardhaman town). Records on file show the store at 713407 (Memari). 713401 is a *delivery* target, not the store address. Confirm which is which before launch, since both go on the site and into Google Business.

2.2 Service geography

Delivery cost in this trade is a function of distance and tonnage. Tiles are heavy, low-value-per-kg freight; a free-delivery-everywhere promise destroys margin. The app must therefore model coverage in explicit zones:

Zone	Coverage	Indicative radius	Delivery
Zone A — Core	Memari, Bardhaman town (713101/713401/713407 and adjacent)	0–25 km	Free above an order threshold
Zone B — District	Kalna, Katwa, Jamalpur, Raina, Galsi, Bhatar	25–60 km	Flat per-trip charge
Zone C — Extended	Durgapur, Asansol, Panagarh, Kanksa (Paschim Bardhaman)	60–130 km	Quoted per order, min order applies
Zone D — Enquiry only	Outside the two districts	—	Enquiry captured; no auto-pricing

PIN code is the gate. The customer enters a PIN early (home screen or first estimate) and the app resolves zone, delivery charge, and whether site measurement is offered. A PIN outside the map should not show a dead end — it should show "we'll call you" and capture the lead.

2.3 Why this market, and why now

- **Construction in tier-3 and semi-rural West Bengal is individual-house-led.** The buyer is a family building or renovating one home, once in 15 years. They are high-anxiety, high-research, low-frequency. They want to see, compare, and be reassured — exactly what a photo catalogue does well.
- **Smartphone-first, desktop-never.** Assume 90%+ of sessions on Android phones over 4G, often mid-range devices. Anything built desktop-first will feel broken.
- **The comparison is not Amazon. It is the shop next door.** The competitor set is other local dealers plus a few Durgapur/Asansol showrooms. None of them has a working catalogue with rates and a calculator. That is the entire opening.
- **Trust is the currency.** Shade variation, breakage in transit, and "the tile looked different in the shop" are the standard complaints. The app must handle these head-on (batch/shade notes, breakage policy, real store photos) rather than hide them.

2.4 Revenue logic

The app is not expected to collect the full order value online. Its measurable contribution is:

1. **Qualified leads with quantity attached** — the owner receives "Rahul, Memari, 713407, 480 sq ft vitrified, ₹52/sq ft range" instead of "ki rate?"
2. **Reduced counter time** — browsing happens before the visit, so the shop conversation starts at selection, not education.
3. **Reach beyond walk-in radius** — Zone B and C buyers who would never have driven to Memari.
4. **Advance/token collection** — a small online advance (₹500–₹5,000) to lock an order and cut no-shows.

3. Goals, Non-Goals & Success Metrics

3.1 Goals

#	Goal	Why it matters
G1	Let any customer see the full range with rates, on a phone, in under 10 seconds from landing	Removes the "what do you have?" phone call
G2	Turn a room description into a box count and an estimated total	The conversion moment; nothing else in the district does this

#	Goal	Why it matters
G3	Deliver every enquiry to the owner's phone with product, quantity, PIN and contact attached	Replaces unqualified calls with workable leads
G4	Let the owner update stock, rates and photos himself in under 2 minutes per product	If updating needs a developer, the catalogue goes stale in a month and the project is dead
G5	Rank on local search for marble/tiles queries in Bardhaman	Free, compounding demand

3.2 Non-goals (explicitly out of scope for v1)

- **Full online payment of order value.** Advance/token only. Nobody pays ₹80,000 for tiles online on first contact in this market.
- **Real-time inventory to the piece.** Stock is modelled as In stock / Low / On order. Piece-level accuracy is a warehouse problem, not an app problem, and faking it destroys trust.
- **A native Android/iOS app.** Mobile web only. An app install is a conversion tax on a once-in-15-years purchase.
- **Marketplace / third-party sellers.** Single-vendor.
- **Installation labour marketplace.** v1 refers a mason list; it does not schedule or bill labour.
- **AI room visualiser at launch.** Phase 3. See §12.

3.3 Success metrics

Primary (the ones the owner should judge the project by):

Metric	Target by month 6	Target by month 12
Qualified enquiries/month (product + quantity + PIN captured)	60	150
Enquiry → visit or site-measure rate	30%	40%
Enquiry → confirmed order rate	12%	18%
Revenue attributable to app-originated leads	₹4–6 L/month	₹12–18 L/month

Secondary (product health):

Metric	Target
Calculator completion rate (started → estimate shown)	> 55%
Median time to first estimate	< 90 seconds
Product page load, 4G, mid-range Android	< 2.5 s LCP
Catalogue freshness (products with rate updated in last 60 days)	> 90%
Bounce on landing	< 45%

Counter-metric (the thing that silently kills it): *quote-to-final-price variance*. If the app estimates ₹41,000 and the shop bills ₹56,000, every future estimate is worthless and the app actively damages trust. Track this per order; target variance within $\pm 8\%$ before delivery and labour, and audit any order beyond $\pm 15\%$.

4. Users & Personas

P1 — Sujata, the homebuilder (primary, ~60% of enquiries)

42, schoolteacher in Memari. Building a two-storey house; her husband handles the site, she handles the choosing. Uses WhatsApp and YouTube fluently; has never completed a purchase on an e-commerce app larger than ₹3,000.

- **Wants:** to see what looks good, know if it fits the budget, and not be cheated on rate.
- **Fears:** buying the wrong quantity; the tile looking different at home; being overcharged because she doesn't know the market rate.
- **Needs from the app:** big photos, honest rates, a number for her room, and a phone number that a human answers.
- **Design implication:** zero jargon. "Living room floor" not "vitrified double-charge". Bengali labels available.

P2 — Sk. Rafiq, the mistri/contractor (~25%, highest value)

35, tiling contractor, runs 3–4 sites at a time across Jamalpur and Raina. Buys repeatedly. Knows exactly what he wants and only needs rate, size availability, and stock.

- **Wants:** rate list, stock status, quantity math, fast reorder, and a trade rate.
- **Hates:** slow pages, being forced through a wizard he doesn't need, marketing copy.

- **Needs from the app:** a dense list/table view, a saved list of past selections, one-tap WhatsApp to the owner, and eventually a logged-in trade rate.
- **Design implication:** a power-user path that bypasses the wizard entirely. He is a *minority of users but a majority of volume* — do not optimise him away.

P3 — Amit, the small builder / promoter (~10%, largest ticket)

38, builds 8–16 flat projects in Bardhaman town. Buys in bulk, negotiates hard, pays in stages.

- **Wants:** bulk pricing conversation, consistency of batch across a whole project, and credit terms.
- **Needs from the app:** essentially a well-organised catalogue plus a "bulk/project enquiry" form that routes straight to the owner. Automated pricing must **not** apply to him; every quote is negotiated.

P4 — Rina, the renovator (~5%)

Re-doing one bathroom or a kitchen dado. Small ticket, high aesthetic interest, most likely to use the visualiser later.

P5 — The owner / admin (internal)

The person who must keep this alive. Runs the shop, is on his phone all day, is not a computer user in the desktop sense.

- **Wants:** to add a product by taking a photo and typing a rate. To see today's enquiries. To mark something out of stock.
- **Absolute requirement:** the admin panel must work **on his phone**, in under 2 minutes per product, with no training. A desktop-only admin CMS will not be used, the catalogue will rot, and the project will fail. This is the single highest-risk requirement in the document.

P6 — Staff (Phase 2)

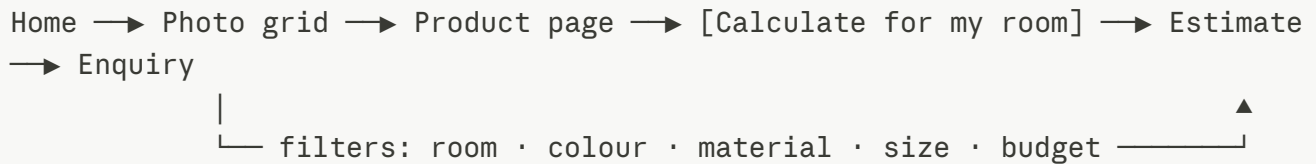
One or two counter staff handling enquiry follow-up, marking quotes sent, and confirming orders. Needs a shared enquiry inbox with assignment and status.

5. Core Product Concept

The home screen offers exactly **two doors**. Not a hero banner, not a carousel of offers — two doors, each a large tap target, plus a phone/WhatsApp button always in reach.

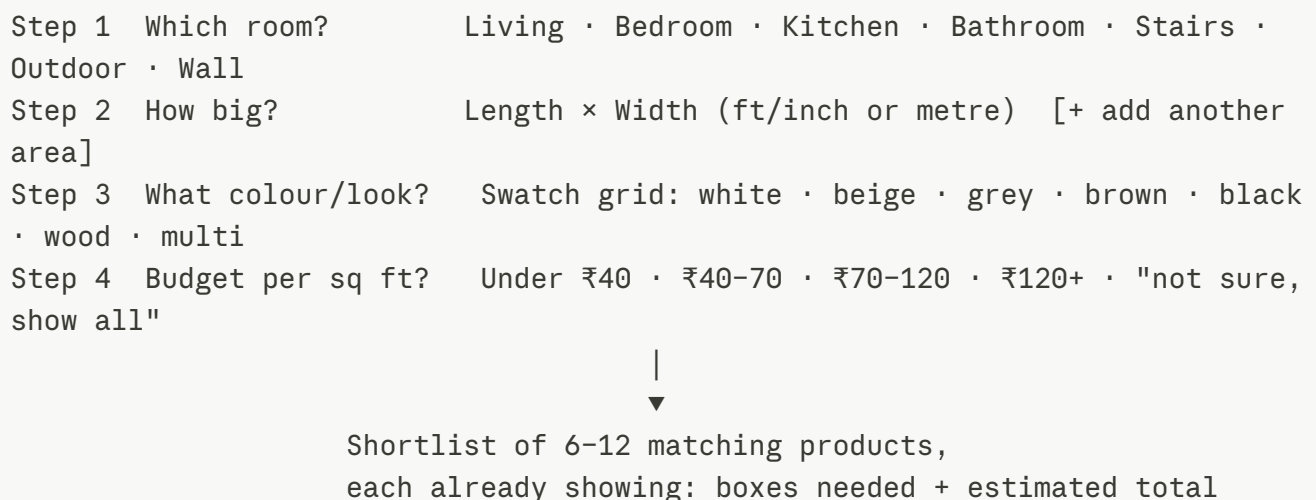
Door A — "Dekhe nin" / Browse the collection

For the customer who wants to look first. Straight into a photo grid of real products with rate per sq ft visible on the tile itself. Filters sit behind one button, not spread across the screen. This is the path for P1 early-stage and P2 always.



Door B — "Amar ghor er jonno" / Tell us your requirement

For the customer who has a room and a budget but no idea what to look at. A 4-step wizard that ends in a *shortlist*, not a single answer.



The crucial difference from a normal filter: **the wizard outputs quantities, not just products**. By the time the customer sees the shortlist, the math is already done for their room, per option. That is the "wow" moment and the reason they send the enquiry.

The two doors converge

Both paths land on the same **Estimate screen**, which is the app's real centre of gravity:

- Selected product(s), with photo
- Area entered, area after wastage
- Boxes required, pieces per box, total sq ft purchased
- Rate applied, material subtotal
- Optional add-ons: skirting, adhesive/cement, grout, cutting charge
- Delivery estimate by PIN zone
- **Estimated total, clearly labelled "Estimate — final rate confirmed by shop"**

- Two actions: **Send enquiry on WhatsApp** (primary) and **Save/Share estimate** (secondary)

Room-level thinking (the differentiator)

An Indian tile purchase is almost never one product. It is "floor for the whole house, different tile for the two bathrooms, something better for the living room". v1.5 should support a **My House** list: several areas, each with its own product and quantity, summing to one estimate. This is what contractors and builders actually need, and no local competitor offers it.

What the customer must never be asked

- To create an account before seeing anything
- To choose between "vitrified", "ceramic", "porcelain" before being shown what they look like
- To enter an email address
- To install an app

6. Functional Requirements

Priority key: **P0** = MVP, must ship. **P1** = Phase 2. **P2** = Phase 3.

6.1 Catalogue & product data

ID	Requirement	Pri
F-1.1	Products organised by category (Marble, Granite, Vitrified, Ceramic Floor, Wall Tile, Outdoor/Parking, Stairs/Riser)	P0
F-1.2	Each product carries: name, category, size, finish, colour family, rate per sq ft, pieces/box, sq ft/box, stock status, photos	P0
F-1.3	Minimum 3 photos per product: flat/close-up, laid-floor context shot, and a scale/edge shot	P0
F-1.4	"Shade may vary by batch" notice on every natural stone product	P0
F-1.5	Related/similar products strip at the bottom of the product page	P1

ID	Requirement	Pri
F-1.6	Tag products as Premium / Value / Economy for budget filtering	P0
F-1.7	Bulk CSV import of catalogue	P1

6.2 Discovery & browse

ID	Requirement	Pri
F-2.1	Photo grid, 2 columns on mobile, image-dominant with name + rate overlaid	P0
F-2.2	Filters in a bottom sheet: room, colour, material, size, finish, budget band, in-stock only	P0
F-2.3	Sort: relevance, rate low→high, rate high→low, newest	P0
F-2.4	Colour filter shown as swatches, not words	P0
F-2.5	Text search with Bengali + Banglish tolerance ("marbel", "tiles", "পাটনা টাইলস", "পাটনা টাইলস")	P1
F-2.6	Compare up to 3 products side by side	P1
F-2.7	Infinite scroll with skeleton placeholders; no pagination	P0

6.3 Requirement wizard (Door B)

ID	Requirement	Pri
F-3.1	4-step wizard: room → dimensions → colour → budget, one question per screen, back always available	P0
F-3.2	Accept feet+inches, feet decimal, and metres; default feet+inches	P0
F-3.3	Support multiple areas in one session ("add another room")	P1
F-3.4	Support irregular rooms as a sum of rectangles ("L-shaped? add two parts")	P1

ID	Requirement	Pri
F-3.5	Every step skippable; skipping widens rather than blocks results	P0
F-3.6	Output is a ranked shortlist with per-product box count and estimated total	P0
F-3.7	Room-type presets suggest sensible defaults (bathroom → anti-skid, small format; parking → heavy-duty)	P1

6.4 Product page

ID	Requirement	Pri
F-4.1	Must fit one mobile screen with no page scrolling (per project UI standard)	P0
F-4.2	Above the fold: photo, name, rate/sq ft, size, stock, and the primary CTA	P0
F-4.3	Secondary detail (finish, pieces/box, application, care) in an expandable sheet, not inline scroll	P0
F-4.4	Primary CTA is "Calculate for my room" , not "Add to cart"	P0
F-4.5	Pinch-zoom full-screen photo viewer	P0
F-4.6	Persistent WhatsApp button that pre-fills the product name and link	P0
F-4.7	"Ask for a sample piece" request	P1

6.5 Estimate & cart

ID	Requirement	Pri
F-5.1	Estimate screen showing area, wastage, boxes, rate, subtotal, add-ons, delivery, total	P0
F-5.2	Every estimate carries a visible disclaimer that the final rate is confirmed by the shop	P0

ID	Requirement	Pri
F-5.3	Editable wastage % (default 8%, range 5–15%) with a plain-language explanation	P0
F-5.4	Optional add-ons: skirting (running ft), adhesive/cement bags, grout, cutting charge	P1
F-5.5	Multi-area "My House" estimate with per-area breakdown and grand total	P1
F-5.6	Share estimate as WhatsApp text and as a one-page PDF	P1
F-5.7	Estimate persists in browser storage for 30 days without login	P0
F-5.8	Unique estimate reference number (e.g. RM-2609-0147) quoted in shop and in the admin panel	P0

6.6 Enquiry & order

ID	Requirement	Pri
F-6.1	Enquiry form: name, phone, PIN, preferred contact time. Nothing else.	P0
F-6.2	Phone OTP verification to stop junk leads	P0
F-6.3	On submit: enquiry saved to admin, WhatsApp message fired to the owner, confirmation SMS to the customer	P0
F-6.4	"Request site measurement" option for Zone A/B	P1
F-6.5	Bulk/project enquiry form routed separately, no auto-pricing	P0
F-6.6	Owner converts enquiry → quote → order in the admin panel, with status visible to the customer via a link	P1
F-6.7	Online advance/token payment (UPI, ₹500–₹5,000) to confirm an order	P1
F-6.8	Full online payment with part-payment schedule	P2

6.7 Trust & content

ID	Requirement	Pri
F-7.1	Store page: real photos, address, map, hours, phone, directions	P0
F-7.2	Clear policy pages: breakage in transit, shade variation, return/exchange, delivery	P0
F-7.3	Customer work gallery — real floors laid in real local homes, with the product tagged	P1
F-7.4	Google reviews embedded	P1
F-7.5	Short guides: how to measure your room, how much wastage to keep, matt vs glossy, anti-skid for bathrooms	P1

6.8 Product variants, stock lots & role-based pricing (added from UI concept review)

A UI concept prepared for this project surfaced real trade requirements not yet captured above. These are now specified:

ID	Requirement	Pri
F-8.1	A product can have multiple variants by thickness and finish (e.g. Absolute Black in 18 mm/20 mm × Polished/Honed/Leather), each with its own rate and stock status	P0
F-8.2	Only finishes actually offered at a given thickness are selectable; others are visibly disabled, not hidden	P0
F-8.3	Natural stone variants may carry a lot note (e.g. "Lot 42 — light veining, 3 slabs left") so shade/veining expectations are set before purchase	P1
F-8.4	Per-variant stock state: In godown / Low stock / Out of stock, each with distinct styling	P0

ID	Requirement	Pri
F-8.5	When a selected variant is out of stock, the calculator is replaced by a "Request restock" flow: a lead-capture sheet (name, phone, requirement) that notifies the owner and tells the customer they'll be called when the lot arrives	P0
F-8.6	Dealer/trade pricing shown as a rate differential once a contractor account is logged in, never as a manual toggle exposed to retail customers	P1
F-8.7	Natural stone technical specs displayed on the product page: water absorption, compressive strength, quarry origin	P1

Design decision needed: the reviewed concept includes a customer-facing "Customer / Dealer" pricing toggle for preview purposes. In the real app this must be replaced by **account-based rate resolution** — a logged-in contractor sees their rate automatically (see Customer.trade_discount_pct, §13). A self-service toggle that lets any visitor see the trade rate defeats the purpose of a trade rate and should not ship.

6.9 Customer order tracking (refines F-6.6)

The reviewed concept also specifies the self-service order tracker referenced in F-6.6. This supersedes that line item with detail:

ID	Requirement	Pri
F-9.1	An "Orders" view, reached without a full account — phone-number lookup is enough — listing Active and Completed orders in tabs	P1
F-9.2	Each order card shows order code, product, total, current status, and expected date at a glance, collapsed by default	P1
F-9.3	Expanding an order shows a step tracker (Order received → Cutting/prep → Dispatched → Out for delivery → Delivered), each step timestamped once reached	P1
F-9.4	Line items, delivery address and payment status (advance paid / balance due / paid in full) shown per order	P1

ID	Requirement	Pri
F-9.5	"Ask about this order" routes straight to WhatsApp with the order code pre-filled	P1
F-9.6	"Order again" on a completed order re-opens the estimate flow pre-filled with the same product and quantity	P2

7. Quantity & Pricing Engine

This section is the technical heart of the product. Get it wrong and the app is worse than useless — it produces confident wrong numbers.

7.1 Area

```
floor_area_sqft = Σ (length_ft × width_ft)    for each rectangle the user enters
```

Inches convert as `ft + in/12`. Walls/dado use the same formula with height in place of width, minus door and window openings if the user supplies them (Phase 2; v1 just notes "deduct openings at the shop").

7.2 Wastage

```
area_with_wastage = floor_area_sqft × (1 + wastage_pct)
```

Situation	Default wastage
Simple rectangular room, large-format tile	8%
Rooms with many cuts / L-shapes / pillars	10-12%
Diagonal or pattern laying	12-15%
Small-format or mosaic	10%
Natural marble/granite slab	12%+ and shop-confirmed

Default is **8%**, user-adjustable, with a one-line explanation: *"Extra tiles for cutting at edges and spare for future repair."* Under-estimating wastage is the worst failure mode here — a customer who runs short mid-job may not get the same batch again.

7.3 Boxes

```
boxes_required = ceil( area_with_wastage / sqft_per_box )
actual_sqft_purchased = boxes_required × sqft_per_box
pieces_total = boxes_required × pieces_per_box
```

`ceil`, never round — you cannot buy 0.6 of a box. The estimate must show **both** the area needed and the area actually purchased, because the gap surprises people at the counter.

Common reference values (to seed the catalogue; confirm against actual stock):

Tile size	Pieces/box	Approx sq ft/box
600 × 600 mm	4	15.50
800 × 800 mm	3	20.67
600 × 1200 mm	2	15.50
300 × 300 mm	11	10.66
300 × 600 mm (wall)	6	11.63
400 × 400 mm	6	10.33

Marble and granite are sold by the sq ft, not by the box. The engine must branch: `pricing_unit = box | sqft | slab`. For slab material, skip the box math, apply wastage, and price on area — and force a "confirm with shop" flag, since slab sizes are irregular.

7.4 Material cost

```
material_cost = actual_sqft_purchased × rate_per_sqft
```

Rate resolution order: customer-specific trade rate (Phase 2) → active promotion → base rate. Rates are stored **exclusive of GST**, with GST shown as a separate line (tiles and marble are commonly at 18% — confirm current slab with the accountant before launch and store it as configuration, never hardcoded).

7.5 Add-ons

```

skirting_cost    = room_perimeter_ft × skirting_rate_per_ft
(perimeter = 2×(L+W))
adhesive_bags    = ceil( area_with_wastage / coverage_sqft_per_bag ) (default
35 sq ft/bag)
grout_kg         = area_with_wastage × grout_factor                      (default
0.10 kg/sq ft)
cutting_charge   = flat, per-order, config

```

All add-ons are **opt-in and clearly separable**. Bundling them silently into the headline number is the fastest way to blow the ±8% variance target.

7.6 Delivery

```

delivery_charge = f(zone, total_weight_kg, order_value)
total_weight_kg ≈ actual_sqft_purchased × weight_per_sqft    (~2.0 kg/sq ft
for 8-10 mm vitrified)

```

Zone A free above threshold; Zone B flat per trip; Zone C quoted, not auto-calculated. Above a configurable tonnage the app should stop calculating and say *"Large order — delivery quoted by the shop"* rather than guess.

7.7 Labour

v1 does **not** price labour. It shows an indicative range as information only ("laying charges in this area are typically ₹22-35/sq ft — arranged separately"), clearly outside the estimate total. Pricing labour the shop doesn't control is a liability.

7.8 The total

```

estimate_total = material_cost
                + add_ons
                + delivery_charge
                + GST

```

Displayed as a breakdown, never as a single opaque number, and always under the label **"Estimate — final rate confirmed by shop"**.

7.9 Worked example (build the test case from this)

Living room 14 ft × 12 ft, 600×600 vitrified at ₹52/sq ft, Zone A:

Step	Value
Floor area	168.00 sq ft
Wastage @ 8%	181.44 sq ft
Sq ft per box	15.50
Boxes (ceil 11.70)	12
Actual purchased	186.00 sq ft
Material @ ₹52	₹9,672
Skirting 52 running ft @ ₹30	₹1,560
Adhesive 6 bags @ ₹380	₹2,280
Subtotal	₹13,512
GST @ 18%	₹2,432
Delivery (Zone A, above threshold)	₹0
Estimate total	₹15,944

This example must exist as an automated regression test. Any change to the engine that moves this number without an explicit reason is a bug.

8. Admin & Operations

8.1 The governing principle

The admin panel is **a phone-first tool for a shopkeeper, not a dashboard for an analyst**. Every screen is designed for one thumb, standing at a counter, possibly with dusty hands. If adding a product takes more than 2 minutes or requires a laptop, the catalogue will go stale within weeks and the entire investment is lost. This is the project's #1 operational risk.

8.2 Catalogue management

ID	Requirement	Pri
A-1.1	Add product from phone: take/upload photo → name → category → size → rate → save. Everything else optional.	P0

ID	Requirement	Pri
A-1.2	Auto image handling: crop, compress, generate thumbnails server-side. Owner never resizes anything.	P0
A-1.3	Edit rate inline from a list view — tap number, type, done	P0
A-1.4	Toggle stock status: In stock / Low / Out of stock / On order	P0
A-1.5	Hide/unhide a product without deleting it	P0
A-1.6	Bulk rate change by percentage across a category (rates move together when a supplier revises)	P1
A-1.7	Duplicate a product as the basis for a similar one	P1

8.3 Enquiry inbox

ID	Requirement	Pri
A-2.1	Chronological list: name, phone, PIN, product, quantity, estimated value, time	P0
A-2.2	One-tap call and one-tap WhatsApp from each enquiry	P0
A-2.3	Status pipeline: New → Contacted → Quoted → Visited → Won → Lost, with a lost-reason picker	P0
A-2.4	Notes field per enquiry	P0
A-2.5	Instant WhatsApp/push alert to the owner on every new enquiry	P0
A-2.6	Assign an enquiry to a staff member	P1
A-2.7	Follow-up reminder after N days with no contact	P1

8.4 Quotes & orders

ID	Requirement	Pri
A-3.1	Convert an enquiry into a formal quote: adjust rate, quantity, delivery; send as PDF/WhatsApp	P1

ID	Requirement	Pri
A-3.2	Quote validity date (rates move; an open-ended quote is a liability)	P1
A-3.3	Convert quote → order with advance recorded	P1
A-3.4	Order status the customer can check by link: Confirmed → Packed → Out for delivery → Delivered	P1
A-3.5	Delivery scheduling with vehicle and driver note	P2

8.5 Operational policies the app must encode

These are business decisions the owner must make; the app has to reflect them, not invent them.

- **Breakage in transit.** Who bears it, and what the customer must do (inspect before the vehicle leaves, photograph, report within X hours). State it plainly on the delivery policy page and at checkout.
- **Shade/batch variation.** Standard trade reality for both tiles and natural stone. Advise buying the full requirement in one batch; say so at the estimate stage, not after delivery.
- **Returns.** Unopened full boxes within N days, restocking fee or not, cut/custom items non-returnable.
- **Minimum order by zone.** Prevents a ₹3,000 order 90 km away.
- **Rate validity.** How long a displayed rate holds. Recommend a footer line: *"Rates updated <date>. Final rate confirmed at the shop."*

8.6 Reporting (light)

A single screen, weekly: enquiries received, conversion, top 10 viewed products, top 10 enquired products, zone distribution. The most commercially useful output is **viewed-but-never-enquired** products — these are either priced wrong or photographed badly, and the owner can act on that the same week.

9. Non-Functional Requirements

9.1 Performance (the hard constraint)

This is an image-heavy app for mid-range Android phones on rural 4G. Performance is not polish here; it is the difference between a sale and a bounce.

ID	Requirement
N-1.1	LCP < 2.5 s on a simulated 4G / mid-tier Android profile
N-1.2	Initial JS bundle < 150 KB gzipped
N-1.3	Grid thumbnails ≤ 40 KB each; full product images ≤ 180 KB, served as WebP/AVIF with JPEG fallback
N-1.4	Lazy-load below the fold; explicit width/height on every image to prevent layout shift (CLS < 0.1)
N-1.5	The app must remain usable on a 2G-equivalent connection, degrading to lower-resolution images
N-1.6	No layout blocked on a third-party script

9.2 Language

ID	Requirement
N-2.1	Bengali and English, switchable, with the choice remembered
N-2.2	Bengali is a first-class translation, not machine output — all trade terms reviewed by the owner
N-2.3	Numbers, rates and dimensions always in Latin numerals even in Bengali text
N-2.4	Search tolerates Banglish transliteration

9.3 Accessibility & usability

ID	Requirement
N-3.1	Minimum tap target 44 × 44 px
N-3.2	Base font 16 px; never below 14 px for content
N-3.3	WCAG AA contrast on all text, including text over photographs
N-3.4	Usable one-handed: primary actions in the lower half of the screen

ID	Requirement
N-3.5	Works with the browser's text-size increased to 150%

9.4 Reliability, security, compliance

ID	Requirement
N-4.1	99.5% uptime target; catalogue served statically so it survives backend downtime
N-4.2	HTTPS everywhere; HSTS
N-4.3	Phone numbers and enquiry data encrypted at rest; access limited to owner and assigned staff
N-4.4	Rate-limit OTP and enquiry submission; bot protection on forms
N-4.5	Daily automated database backup with a tested restore procedure
N-4.6	Privacy policy and terms published; explicit consent line before phone submission (DPDP Act)
N-4.7	GST number and legal business details in the footer

9.5 SEO & discoverability

Organic local search will out-perform paid advertising here over a 12-month horizon.

ID	Requirement
N-5.1	Server-rendered product and category pages with unique titles and meta descriptions
N-5.2	Product schema.org markup with price and availability
N-5.3	Location landing pages: "Tiles shop in Memari", "Marble dealer in Bardhaman", "Tiles in Kalna"
N-5.4	Google Business Profile linked, with the same NAP (name, address, phone) as the site
N-5.5	Sitemap, robots.txt, canonical URLs

ID	Requirement
N-5.6	Every product page shareable on WhatsApp with a correct preview image and title

9.6 Analytics

GA4 or a lightweight privacy-friendly alternative, tracking: product views, filter use, wizard start/step/complete, estimate generated, WhatsApp click, enquiry submit. Funnel visibility matters more than page views — the question to answer weekly is *where people drop out between photo and enquiry*.

10. Design Standards

10.1 Binding project standard

Every screen in this product is built to the **roman-marbel-ui** standard, and its pre-ship checklist is run on every screen before that screen is considered done. No exceptions, no "we'll align it later". Where anything in this PRD appears to conflict with that standard, the standard wins and the PRD gets amended.

10.2 Hard layout rules

- **Mobile web, not a native app.** Design at 360–430 px width first; desktop is a widened variant, never the design target.
- **Product pages fit one mobile screen with no page scrolling.** Everything that doesn't fit goes into an expandable sheet or a secondary view. This is a constraint on *how much a product page may claim*, and it forces the right discipline: photo, name, rate, size, stock, one CTA.
- **Simple over feature-dense.** When a screen can be split or simplified, split or simplify it. A screen that requires explanation has failed.

10.3 Visual direction

The product being sold is a surface. The interface should behave like good gallery lighting: neutral, quiet, and entirely deferential to the material.

- **Let the photography carry the page.** Chrome stays neutral; saturated brand colour anywhere near a product image distorts how the customer reads the tile's shade — and shade is the purchase decision.
- **Generous whitespace.** Crowding reads as cheap, and this shop sells a premium-feeling category.

- **One primary action per screen**, visually unambiguous, thumb-reachable.
- **Photo consistency matters more than photo styling.** Same lighting, same angle, same white balance across the catalogue. An inconsistent catalogue looks untrustworthy regardless of how good the individual shots are. Budget for a single proper shoot day rather than accumulating phone snaps over months.

10.4 Content tone

Plain, respectful, no hard sell. The customer is spending a large sum on something permanent in their home. Confidence and clarity sell here; urgency banners and countdown timers do not, and will actively damage trust in a market built on personal reputation.

10.5 Screens to design (MVP)

#	Screen	Notes
1	Home — two doors	Plus persistent call/WhatsApp
2	Photo grid	2-col, filter button, sort
3	Filter sheet	Bottom sheet, swatch-based colour
4	Product page	One screen, no scroll
5	Photo viewer	Full-screen, pinch-zoom
6	Wizard steps 1–4	One question per screen
7	Shortlist	Results with boxes + total
8	Estimate	The conversion screen
9	Enquiry form	Name, phone, PIN, time
10	Confirmation	Reference number, what happens next
11	Store / contact	Map, hours, directions
12	Policies	Delivery, breakage, returns
13–18	Admin: login, product list, add/edit product, enquiry inbox, enquiry detail, settings	Phone-first
19	Orders / order tracking	Active/Completed tabs, step tracker per order — see F-9.1–F-9.6

11. Technical Architecture

11.1 Guiding constraints

One shop, one admin user, a few thousand catalogue items at most, and perhaps 5,000–20,000 sessions a month at maturity. This is a **small-data, image-heavy, read-mostly** application. The engineering risk is not scale — it is over-engineering. A microservices architecture here is malpractice; so is a hand-rolled PHP admin that nobody can maintain.

Second constraint: **the owner must not need a developer for day-to-day operation.** Anything that requires a deploy to change a rate is a design failure.

11.2 Recommended stack

Layer	Choice	Reasoning
Frontend	Next.js (App Router), TypeScript, Tailwind	SSR/ISR gives the SEO that matters; static catalogue pages are fast and cheap
Rendering	ISR for catalogue, SSR for search/estimate, client-side only for the wizard	Catalogue changes rarely; regenerate on publish
Backend	Next.js API routes / a thin Node service	One deployable; no cross-service latency to pay for
Database	PostgreSQL (managed — Supabase or Neon)	Relational fits the data; managed removes ops burden
Images	Cloudinary or ImageKit	Automatic format negotiation, resizing, CDN. Do not build this.
Auth	Phone OTP via an Indian SMS provider (MSG91 / Fast2SMS); admin behind a separate credentialed login	Phone is the identity that exists here
Hosting	Vercel (frontend) + managed DB	Zero-ops, free-to-cheap at this volume
Messaging	WhatsApp deep links in v1; WhatsApp Business Cloud API in Phase 2	Deep links cost nothing and work today
Payments	Razorpay (UPI-first)	Best UPI coverage; needed only for advances
Analytics	GA4 or Plausible	—

11.3 Why not the obvious alternatives

- **Shopify / WooCommerce.** Both assume box-quantity retail. Neither models sq ft → wastage → boxes, room-based multi-area estimates, or enquiry-led selling without heavy plugin work that becomes its own maintenance burden. The calculator *is* the product; building around a platform that fights it is the wrong trade.
- **A native app.** Install friction on a once-in-15-years purchase. Revisit only if repeat contractor usage becomes significant.
- **Headless CMS.** Another system for the owner to learn. Product data belongs in the app's own database with a phone-first admin.

11.4 Image pipeline (the highest-leverage engineering decision)

Owner uploads from phone

- ↓ client-side downscale to max 2000 px before upload (saves rural upload time)
- ↓ store original in image CDN
- ↓ derive: 400 px thumb (WebP/AVIF), 800 px product, 1600 px zoom
- ↓ serve via srcset with explicit dimensions, lazy-loaded

Every image served through the CDN's transformation URL. No original-size images ever reach a customer's phone. This single decision determines whether the app feels fast or broken.

11.5 Environments & delivery

- Three environments: local, staging, production. Staging seeded with real catalogue data.
- Git-based deploys with preview URLs per branch — the owner reviews on his phone before anything goes live.
- Automated tests: the §7.9 worked example plus a table of calculator cases (metric/imperial, slab vs box, zero/negative input, absurd input like 9999 ft).
- Error tracking (Sentry) from day one. Errors on the estimate screen must page someone.

11.6 Indicative cost envelope (monthly, INR)

Item	Cost
Hosting (Vercel hobby/pro)	₹0–1,700
Managed Postgres	₹0–2,000

Item	Cost
Image CDN	₹0–2,000
SMS/OTP	₹300–1,200
Domain	₹100
Running total	₹500–7,000/month

Build cost is dominated by design and photography, not infrastructure. Budget a proper product photography day — it will return more than any feature on the P1 list.

11.7 Team

MVP is deliverable by one strong full-stack engineer plus a designer, over 8–10 weeks, with the owner giving 2–3 hours a week for catalogue and policy decisions. Ongoing, it needs roughly 0.2 FTE for maintenance. The owner's weekly time is a real dependency — if it isn't committed, the timeline slips regardless of engineering capacity.

12. AI/ML Roadmap

12.1 The discipline

Every item below is ranked by **value per unit of engineering risk**, not by how impressive it sounds. The most tempting feature (room visualisation) is deliberately last, because it is the most expensive, the most fragile, and the least proven to convert in this segment. Shipping it before the catalogue is complete and the calculator is trusted would be a serious sequencing error.

12.2 Tier 1 — ship in Phase 2, low risk, real value

Feature	How	Value
Colour extraction & auto-tagging	Run k-means on each product photo at upload; derive dominant colour family and store it	Removes manual tagging from the owner's workflow; powers the swatch filter reliably
"Similar looking" recommendations	Perceptual embedding (CLIP or a small pretrained vision model) → nearest neighbours by visual similarity	Solves the real customer behaviour: "I like this one but cheaper". Purely visual, no training data needed

Feature	How	Value
Search that tolerates how people actually type	Fuzzy matching + a Banglish transliteration map + synonym table	"bathrum tils", "kalo marbel" must return results
Lead scoring	Simple logistic model or even weighted rules over: estimate value, zone, wizard completion, repeat visits	Owner calls the ₹80,000 Zone-A lead before the ₹6,000 Zone-C one. Highest ROI item on this list.

Note that three of these four need no training data at all, and the fourth needs only a few hundred labelled enquiry outcomes, which the app generates naturally.

12.3 Tier 2 — Phase 3, moderate risk

Feature	How	Caveats
Photo-to-product matching	Customer uploads a photo of a tile they saw; embedding search returns closest in-catalogue matches	Works well for tiles, poorly for marble (shade variance). Must present results as "closest we have", never "this is it"
Guided style recommendation	LLM over the catalogue: "south-facing small bedroom, wooden look, under ₹60" → shortlist with reasoning	Must be grounded strictly in actual catalogue data with a rates-and-stock check, or it will confidently recommend things the shop doesn't sell
Demand forecasting	Seasonality + enquiry trend per category → what to stock before the festive/construction season	Needs 12+ months of data before it is worth anything. Do not attempt earlier.

12.4 Tier 3 — the room visualiser

The feature everyone asks for first. Here is what it actually costs.

Option A — Flat perspective overlay (recommended first attempt). Customer photographs their room; app detects the floor plane and applies a perspective-warped tile texture. Doable with a segmentation model plus a homography transform. Works acceptably on clear, well-lit, empty floors. Fails on cluttered rooms, poor light, and odd angles — which describes most real under-construction sites.

Option B — Curated room templates (recommended actual approach). Pre-shot photographs of 8–10 typical room types with the floor masked out; the customer picks a template close to their room and sees their chosen tile applied. Not personalised, but it is reliable, fast, cheap, and answers the real question — *what does this look like across a whole floor rather than as a 4-inch sample* — which is what customers are actually asking.

Option C — Generative room rendering. Diffusion-based. Impressive in a demo, but it invents details, cannot guarantee the tile shown is the tile sold, and creates a genuine misrepresentation risk in a purchase this large. **Not recommended.** If it ever ships, every output must be watermarked as an illustration.

Recommendation: build Option B in Phase 3. Evaluate Option A only after B proves customers use visualisation at all. Ship neither until the catalogue is complete, the calculator is trusted, and enquiry volume justifies the spend.

12.5 The honest summary

If only one AI feature is ever built, build **lead scoring** — it directly increases revenue by ordering the owner's follow-up calls, costs a fraction of the visualiser, and cannot embarrass the business in front of a customer.

13. Data Model

Core entities and the fields that matter. Everything is soft-deleted; nothing in a retail catalogue should ever be hard-deleted, because old estimates reference it.

Product

```
id · sku · name_en · name_bn · slug
category_id · material (marble|granite|vitrified|ceramic|other)
size_mm (600x600) · finish (glossy|matt|rustic|polished|anti-skid)
colour_family[] · colour_hex_primary (auto-extracted)
pricing_unit (box|sqft|slab)
rate_per_sqft · mrp_per_sqft · trade_rate_per_sqft
pieces_per_box · sqft_per_box · weight_per_sqft
stock_status (in|low|out|on_order)
budget_band (economy|value|premium)
application[] (floor|wall|bathroom|kitchen|outdoor|stairs|parking)
is_published · is_featured · shade_variation_note
rate_updated_at · created_at · updated_at
```

ProductVariant — added after UI concept review; a product's sellable thickness/finish combinations

```
id · product_id · thickness_mm · finish
(polished|honed|leather|matt|glossy|anti_skid)
retail_rate_per_sqft · dealer_rate_per_sqft
stock_status (in|low|out) · lot_note · water_absorption_pct ·
compressive_strength_mpa · quarry_origin
created_at · updated_at
```

ProductImage — id · product_id · cdn_public_id · type (flat|context|scale) ·
sort_order · alt_en · alt_bn

Category — id · name_en · name_bn · slug · parent_id · sort_order · icon

Estimate — the app's most important record

```
id · reference_no (RM-2609-0147) · session_id · customer_id (nullable)
pin_code · zone
wastage_pct · subtotal · addons_total · delivery_charge · gst_amount · total
source (browse|wizard) · created_at · expires_at
```

EstimateLine — one per area

```
id · estimate_id · product_id · area_label ("Living room")
length_ft · width_ft · area_sqft · area_with_wastage
boxes · actual_sqft · rate_applied · line_total
skirting_ft · adhesive_bags · grout_kg
```

Enquiry

```
id · estimate_id (nullable) · name · phone · phone_verified · pin_code · zone
preferred_contact_time · message · type (standard|bulk|sample|site_visit)
status (new|contacted|quoted|visited|won|lost) · lost_reason
assigned_to · lead_score · source_page · created_at · last_contacted_at
```

Order (Phase 2) — id · enquiry_id · customer_id · quoted_total · final_total ·
advance_paid · payment_status · delivery_status · delivery_date ·
vehicle_note

Customer — id · phone · name · pin_code · type (retail|contractor|builder) ·
trade_discount_pct · created_at

Zone — id · name · pin_codes[] · delivery_flat_charge · free_above ·
min_order · max_auto_weight_kg · site_visit_available

Setting — key · value for GST rate, default wastage, adhesive coverage, grout factor,
rate-validity text. Every business constant lives here, never in code.

Design notes

- `rate_updated_at` on Product drives the catalogue-freshness metric in §3.3 and the "rates updated on" footer.
- `EstimateLine` stores the **rate applied at the time**, not a foreign key lookup. When rates change, old estimates must still show what the customer was told.
- `Estimate` survives without a customer — anonymous estimates are the norm, and forcing identity earlier loses the lead.

14. Release Plan

Phase 0 — Foundation (2 weeks, mostly not engineering)

This phase is where the project is actually won or lost, and it is the one most teams skip.

- Product photography day: consistent lighting, flat + context + scale shot for every SKU
- Catalogue data entry: 80–150 products with real sizes, real rates, real box counts
- Owner decisions: delivery zones, charges, thresholds, breakage policy, return policy, GST treatment
- Design system to the roman-marbel-ui standard; key screens designed and reviewed on the owner's own phone

Gate: no engineering on the customer app starts until the catalogue data and photography exist. Building a beautiful shell around an empty catalogue is the classic failure mode.

Phase 1 — MVP (6–8 weeks)

All P0 items from §6.

Week	Focus
1–2	Schema, admin: add/edit product, image pipeline, owner starts loading catalogue in parallel
3–4	Catalogue: grid, filters, product page, photo viewer
5	Calculator engine + estimate screen, with the §7.9 test suite
6	Wizard, enquiry flow, OTP, WhatsApp routing, admin inbox

Week	Focus
7	Bengali, SEO, performance pass, policy pages, analytics
8	UAT with 8-10 real customers at the counter, fixes, launch

Launch definition of done: the owner has added 5 products himself on his phone without help; a real customer has completed browse → estimate → enquiry unaided; the §7.9 test passes; LCP is under 2.5 s on a real mid-range Android over 4G.

Phase 2 — Depth (8-10 weeks, starting ~1 month after launch)

Only after at least 4 weeks of live enquiry data.

- Multi-area "My House" estimates
- Quote → order → advance payment (Razorpay UPI)
- Order status tracking by link
- Work gallery, guides, reviews
- Compare, sample requests, site-measurement booking
- Tier 1 AI: colour auto-tagging, similar-looking recommendations, lead scoring
- Staff accounts and assignment

Phase 3 — Differentiation (evaluate at month 9)

Gated on Phase 2 metrics being met, not on the calendar.

- Room visualiser, Option B (curated templates)
- Photo-to-product matching
- Contractor accounts with trade rates and reorder
- Demand forecasting once 12 months of data exists

Sequencing principle

Each phase must pay for the next. If Phase 1 does not produce 40+ qualified enquiries a month by month 3, the correct response is to fix the catalogue, photography and rates — not to build Phase 2 features. More features do not fix a catalogue nobody trusts.

15. Risks, Assumptions & Open Questions

15.1 Risks

#	Risk	Impact	Mitigation
R1	Catalogue goes stale. Owner stops updating rates and stock after the first month.	Fatal	Phone-first admin, 2-minute add flow, freshness metric on the admin home, weekly reminder
R2	Estimate variance. App says ₹41k, shop bills ₹56k.	Fatal to trust	Clear "estimate" labelling, separated add-ons, $\pm 8\%$ tracking, audit every order beyond $\pm 15\%$
R3	Rate transparency exposes margin. Competitors and customers both see the rate card.	High	Publish honest rates; compete on selection, trust and service. Attempting to hide rates defeats the app's purpose. Consider MRP + "best rate at shop" framing if needed.
R4	Photos misrepresent shade. Customer rejects delivery.	High	Consistent shoot conditions, explicit shade-variation notes, physical sample offer, no filters or enhancement on product photos
R5	Low traffic after launch. Built it; nobody came.	High	SEO from day one, Google Business Profile, WhatsApp status and local groups, printed QR code at the shop counter and on the delivery vehicle
R6	Enquiries not followed up. Leads rot in the inbox.	High	Instant WhatsApp alert, status pipeline, follow-up reminders, weekly review with the owner
R7	Heavy/fragile logistics. Breakage disputes.	Medium	Written policy, inspection-at-delivery process, photo documentation
R8	Owner time not available. Catalogue and decisions stall.	Medium	Phase 0 gate; make the time commitment explicit before the project starts

#	Risk	Impact	Mitigation
R9	Over-building AI too early. Budget spent on a visualiser instead of the catalogue.	Medium	The phase gates in §14 exist precisely for this
R10	Fabricated activity signals. A reviewed UI concept includes a live "active buyers online" counter, a "people viewing this" badge, and periodic "someone just ordered" toasts — all simulated with random increments and canned messages, not real data.	High — reputational	Do not ship fabricated social proof. In a small district where the owner's name is the brand, a customer noticing fake numbers (trivial — they're static/random) does lasting damage. Either wire these to real events (actual page views, actual recent orders) or remove the feature entirely for v1.

15.2 Assumptions

- The shop can maintain rates within roughly $\pm 8\%$ of what it displays.
- The owner will commit 2–3 hours weekly, and is willing to publish rates publicly.
- Existing stock is broad enough to make a catalogue worth browsing (target: 80+ distinct SKUs).
- WhatsApp is the acceptable primary channel for both the owner and customers.
- Delivery capacity exists (own or hired vehicle) for the zones defined in §2.2.

15.3 Open questions for the owner

These block specific sections and need answers before or during Phase 0.

#	Question	Blocks
Q1	Is the store address 713407 (Memari) with 713401 as a delivery target, or has the store location changed?	§2.1, SEO, Google Business
Q2	Are displayed rates inclusive or exclusive of GST, and what rate applies to each category?	§7.4
Q3	Exact delivery charges and free-delivery thresholds per zone	§2.2, §7.6
Q4	Breakage-in-transit policy — who bears it, and under what conditions?	§8.5

#	Question	Blocks
Q5	Return/exchange policy for unopened boxes	§8.5
Q6	Is a contractor/trade rate offered, and how is eligibility decided?	§7.4, Phase 3
Q7	How many distinct SKUs are currently stocked?	Phase 0 scope
Q8	Does the shop arrange installation labour, or only refer masons?	§7.7
Q9	Is site measurement offered, in which zones, and is it chargeable?	F-6.4
Q10	Which UPI/bank account receives online advances, and who reconciles them?	F-6.7
Q11	Are GST registration and trade licence details available for the footer?	§9.4
Q12	Who answers the enquiry calls — the owner alone, or staff?	§8.3, P6

15.4 Document control

Field	Value
Version	1.0 (Draft)
Status	For owner review
Next review	After Q1–Q12 are answered
Change process	Amend this doc directly; comment inline for anything contested

A closing note on sequencing. The most common way a project like this fails is not bad engineering. It is a beautiful app wrapped around thirty products with inconsistent photos and rates last updated in March. The catalogue, the photography and the owner's two hours a week are the product. Everything in §11 and §12 is scaffolding around them.